

Chapter 7

Solving Nonlinear Least-Squares Problems

7.1 Introduction

This chapter is devoted to solving the following nonlinear least-squares problems:

$$\min_{x \in R^n} f(x) = \frac{1}{2} r(x)^T r(x) = \frac{1}{2} \sum_{i=1}^m [r_i(x)]^2, \quad m \geq n \quad (7.1.1)$$

where $r : R^n \rightarrow R^m$ is a nonlinear function of x . If $r(x)$ is a linear function, the problem (7.1.1) is the linear least-squares problem.

Nonlinear least-squares problem (7.1.1) can be regarded as a special case for unconstrained minimization with a special structure. This problem can also be interpreted as solving the system of m nonlinear equations

$$r_i(x) = 0, \quad i = 1, 2, \dots, m, \quad (7.1.2)$$

where $r_i(x)$ is called the residual function. When $m > n$, the system is called over-determined, and when $m = n$ the system is well-determined.

Nonlinear least-squares problems have wide applications in data fitting, parameter estimation, function approximation, and others. For example, suppose we are given the data $(t_1, y_1), (t_2, y_2), \dots, (t_m, y_m)$ and want to fit a function $\phi(t, x)$ which is a nonlinear function of x . We want to choose x such that the function $\phi(t, x)$ fits the data as well as possible in the sense of

minimizing the sum of the squares of the residual,

$$\min \sum_{i=1}^m [r_i(x)]^2 \quad (7.1.3)$$

where

$$r_i(x) = \phi(t_i, x) - y_i, \quad i = 1, \dots, m \quad (7.1.4)$$

are the residual. Usually, $m \gg n$. So, we obtain the problem (7.1.1). For solving nonlinear least-squares problem, we usually use Newton's method to solve the resulting system of the normal equations. However, it is expensive, and the normal equations tend easily to be ill-conditioned. Note that the problem (7.1.1) has special structure which inspires some special methods. In this chapter, we shall give some effective and special methods for solving nonlinear least-squares problem (7.1.1).

Let $J(x)$ be the Jacobian of $r(x)$,

$$J(x) = \begin{bmatrix} \frac{\partial r_1}{\partial x_1}(x) & \frac{\partial r_1}{\partial x_2}(x) & \cdots & \frac{\partial r_1}{\partial x_n}(x) \\ \frac{\partial r_2}{\partial x_1}(x) & \frac{\partial r_2}{\partial x_2}(x) & \cdots & \frac{\partial r_2}{\partial x_n}(x) \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial r_m}{\partial x_1}(x) & \frac{\partial r_m}{\partial x_2}(x) & \cdots & \frac{\partial r_m}{\partial x_n}(x) \end{bmatrix}. \quad (7.1.5)$$

Then the gradient of $f(x)$ is

$$g(x) = \sum_{i=1}^m r_i(x) \nabla r_i(x) = J(x)^T r(x) \quad (7.1.6)$$

and the Hessian is

$$\begin{aligned} G(x) &= \sum_{i=1}^m (\nabla r_i(x) \nabla r_i(x)^T + r_i(x) \nabla^2 r_i(x)) \\ &= J(x)^T J(x) + S(x), \end{aligned} \quad (7.1.7)$$

where

$$S(x) = \sum_{i=1}^m r_i(x) \nabla^2 r_i(x). \quad (7.1.8)$$

Therefore, the quadratic model of the objective function $f(x)$ is

$$\begin{aligned} q^{(k)}(x) &= f(x_k) + g(x_k)^T (x - x_k) + \frac{1}{2} (x - x_k)^T G(x_k) (x - x_k) \\ &= \frac{1}{2} r(x_k)^T r(x_k) + (J(x_k)^T r(x_k))^T (x - x_k) \\ &\quad + \frac{1}{2} (x - x_k)^T (J(x_k)^T J(x_k) + S(x_k)) (x - x_k). \end{aligned} \quad (7.1.9)$$

Then we have Newton's method for (7.1.1),

$$x_{k+1} = x_k - (J(x_k)^T J(x_k) + S(x_k))^{-1} J(x_k)^T r(x_k). \quad (7.1.10)$$

We have seen in Chapter 3 that, under standard assumptions, the iteration (7.1.10) is locally quadratically convergent. However, the main disadvantage of the above Newton's method is that the second-order term $S(x)$ in the Hessian $G(x)$ is difficult or expensive to compute. It is also not suitable to use a secant approximation of the whole of $G(x)$, because $J(x)$ and furthermore the first-order term $J(x)^T J(x)$ in $G(x)$ are available when we compute the gradient $g(x)$. Hence, for reducing the computation, it may be reasonable and effective to either neglect $S(x)$ or use first-order derivative information to approximate $S(x)$. Notice from (7.1.8) that when $r_i(x)$ approaches zero or closes to a linear function, in which case $\nabla^2 r_i(x)$ approaches zero, $S(x)$ is small and can be neglected. We call this case a small residual problem, otherwise, a large residual problem.

7.2 Gauss-Newton Method

In this section, we discuss the Gauss-Newton method, which is obtained by neglecting the second-order term $S(x)$ of $G(x)$ in the quadratic model (7.1.9). So, (7.1.9) becomes

$$\begin{aligned} \bar{q}^{(k)}(x) &= \frac{1}{2} r(x_k)^T r(x_k) + (J(x_k)^T r(x_k))^T (x - x_k) \\ &\quad + \frac{1}{2} (x - x_k)^T (J(x_k)^T J(x_k)) (x - x_k). \end{aligned} \quad (7.2.1)$$

Hence (7.1.10) becomes

$$x_{k+1} = x_k + s_k = x_k - (J(x_k)^T J(x_k))^{-1} J(x_k)^T r(x_k). \quad (7.2.2)$$

To make the iteration well-defined, it is required that Jacobian matrix $J(x_k)$ has full column rank. The following is the Gauss-Newton algorithm.

Algorithm 7.2.1 (Gauss-Newton Method)

Step 0. Given $x_0, \epsilon > 0, k := 0$.

Step 1. If $\|g_k\| \leq \epsilon$, stop.

Step 2 Solve

$$J(x_k)^T J(x_k) s = -J(x_k)^T r(x_k) \quad \text{for } s_k. \quad (7.2.3)$$

Step 3. Set $x_{k+1} = x_k + s_k$, $k := k + 1$. Go to Step 1. $\square$

Obviously, whenever $J(x_k)$ has full rank and the gradient $g(x_k)$ is nonzero, the search direction s_k is a descent direction for f , because

$$s_k^T \nabla f(x_k) = s_k^T J(x_k)^T r(x_k) = -s_k^T J(x_k)^T J(x_k) s_k \leq 0.$$

The final inequality is strict unless $J(x_k)^T s_k = 0$, which is equivalent to $J(x_k)^T r(x_k) = g(x_k) = 0$.

Equation (7.2.3) is said to be the Gauss-Newton equation. Obviously, by comparing (7.2.3) and (7.1.10), we find that the difference between Gauss-Newton method and Newton method is that the first-order term $J(x_k)^T J(x_k)$ is used to replace the Hessian $G(x_k)$.

Note that Step 2 in Algorithm 7.2.1 is just analogous to the normal equations of linear least-squares problem. Besides, the model (7.2.1) is equivalent to considering the affine model of $r(x)$ near x_k ,

$$\bar{M}_k = r(x_k) + J(x_k)(x - x_k), \quad (7.2.4)$$

and solve the linear least-squares problem

$$\min \frac{1}{2} \|\bar{M}_k(x)\|^2. \quad (7.2.5)$$

These two observations expose that Gauss-Newton method, in fact, is a linearization method for nonlinear least-squares problem. From (7.2.2), we see that Gauss-Newton method has some advantages in that it only requires the first-order derivative information of the residual function $r(x)$, and that $J(x)^T J(x)$ is at least positive semi-definite.

Since Newton's method, under the standard assumptions, is locally and quadratically convergent, the success of Gauss-Newton method will depend on the importance of the neglected second-order term $S(x)$ in $G(x)$. The following theorem shows:

1. if $S(x^*) = 0$, the Gauss-Newton method is quadratically convergent;
2. if $S(x^*)$ is small relative to $J(x^*)^T J(x^*)$, the Gauss-Newton method is locally Q -linearly convergent;

3. if $S(x^*)$ is too large, the Gauss-Newton method will not be convergent.

The proofs of the following theorem are similar to that of Theorem 3.2.2 for Newton's method. The different proofs given in Theorem 7.2.2 and Theorem 7.2.3 are helpful to study and understand the convergence theorems of various iterative methods.

Theorem 7.2.2 *Let $f : R^n \rightarrow R$ and $f \in C^2$. Assume that x^* is the local minimizer of the nonlinear least-squares problem (7.1.1) and $J(x^*)^T J(x^*)$ is positive definite. Assume also that the sequence $\{x_k\}$ generated by Algorithm 7.2.1 converges to x^* . Then, if $G(x)$ and $(J(x)^T J(x))^{-1}$ are Lipschitz continuous in the neighborhood of x^* , we have*

$$\|x_{k+1} - x^*\| \leq \|(J(x^*)^T J(x^*))^{-1}\| \|S(x^*)\| \|x_k - x^*\| + O(\|x_k - x^*\|^2). \quad (7.2.6)$$

Proof. Since $G(x)$ is Lipschitz continuous, $J(x)^T J(x)$ and $S(x)$ are also Lipschitz continuous. Hence, there exist $\alpha, \beta, \gamma > 0$, such that for any x, y in the neighborhood of x^* , we have

$$\|J(x)^T J(x) - J(y)^T J(y)\| \leq \alpha \|x - y\|, \quad (7.2.7)$$

$$\|S(x) - S(y)\| \leq \beta \|x - y\|, \quad (7.2.8)$$

$$\|(J(x)^T J(x))^{-1} - (J(y)^T J(y))^{-1}\| \leq \gamma \|x - y\|, \quad (7.2.9)$$

(see Exercise).

Since $f \in C^2$ and $G(x)$ is Lipschitz continuous, then we have

$$g(x_k + s) = g(x_k) + G(x_k)s + O(\|s\|^2). \quad (7.2.10)$$

Let $h_k = x_k - x^*$ and $s = -h_k$. We can deduce that

$$0 = g(x^*) = g(x_k) - G(x_k)h_k + O(\|h_k\|^2). \quad (7.2.11)$$

Substituting (7.1.6) and (7.1.7) into (7.2.11) gives

$$J(x_k)^T r(x_k) - (J(x_k)^T J(x_k) + S(x_k))h_k + O(\|h_k\|^2) = 0. \quad (7.2.12)$$

Assume that x_k is in a neighborhood of x^* . From Theorem 1.2.5, it follows that for k sufficiently large, $J(x_k)^T J(x_k)$ is positive definite, and hence $(J(x_k)^T J(x_k))^{-1}$ is bounded above and

$$\|(J(x_k)^T J(x_k))^{-1}\| \leq 2\|(J(x^*)^T J(x^*))^{-1}\|. \quad (7.2.13)$$

Then, multiplying (7.2.12) by $(J(x_k)^T J(x_k))^{-1}$ yields that

$$-s_k - h_k - (J(x_k)^T J(x_k))^{-1} S(x_k) h_k + O(\|h_k\|^2) = 0. \quad (7.2.14)$$

Note that $s_k + h_k = x_{k+1} - x^* = h_{k+1}$, the above equality can be written as

$$\begin{aligned} & -h_{k+1} - (J(x^*)^T J(x^*))^{-1} S(x^*) - (J(x_k)^T J(x_k))^{-1} (S(x_k) - S(x^*)) h_k \\ & - [(J(x_k)^T J(x_k))^{-1} - (J(x^*)^T J(x^*))^{-1}] S(x^*) h_k + O(\|h_k\|^2) \\ & = 0. \end{aligned} \quad (7.2.15)$$

Taking the norm and using (7.2.8)–(7.2.9) and (7.2.13) give the result (7.2.6). $\square$

Theorem 7.2.3 *Let $f : D \subset R^n \rightarrow R$ and $f \in C^2(D)$, where D is an open convex set. Let $J(x)$ be Lipschitz continuous on D , i.e.,*

$$\|J(x) - J(y)\|_2 \leq \gamma \|x - y\|_2, \quad \forall x, y \in D, \quad (7.2.16)$$

and $\|J(x)\|_2 \leq \alpha, \forall x \in D$. Assume that there exist $x^ \in D$ and $\lambda, \sigma \geq 0$ such that $J(x^*)^T r(x^*) = 0$, λ is the smallest eigenvalue of $J(x^*)^T J(x^*)$, and*

$$\|(J(x) - J(x^*))^T r(x^*)\|_2 \leq \sigma \|x - x^*\|_2, \quad \forall x \in D. \quad (7.2.17)$$

If $\sigma < \lambda$, then, for any $c \in (1, \lambda/\sigma)$, there exists $\epsilon > 0$ such that for all $x_0 \in N(x^, \epsilon)$, the sequence generated by Gauss-Newton Algorithm 7.2.1 is well-defined, converges to x^* , and satisfies*

$$\|x_{k+1} - x^*\|_2 \leq \frac{c\sigma}{\lambda} \|x_k - x^*\|_2 + \frac{c\alpha\gamma}{2\lambda} \|x_k - x^*\|_2^2 \quad (7.2.18)$$

and

$$\|x_{k+1} - x^*\|_2 \leq \frac{c\sigma + \lambda}{2\lambda} \|x_k - x^*\|_2 < \|x_k - x^*\|_2. \quad (7.2.19)$$

Proof. By induction. For convenience, let J_0, r_0, r^* denote $J(x_0), r(x_0)$ and $r(x^*)$. From Theorem 1.2.5, it follows that there exists $\epsilon_1 > 0$ such that $J_0^T J_0$ is nonsingular and satisfies

$$\|(J_0^T J_0)^{-1}\| \leq c/\lambda, \quad \text{for } x_0 \in N(x^*, \epsilon_1). \quad (7.2.20)$$

Let

$$\epsilon = \min \left\{ \epsilon_1, \frac{\lambda - c\sigma}{c\alpha\gamma} \right\}, \quad (7.2.21)$$

where γ is the Lipschitz constant defined in (7.2.16). Then, x_1 is well-defined at the first iteration, and we have

$$\begin{aligned} x_1 - x^* &= x_0 - x^* - (J_0^T J_0)^{-1} J_0^T r_0 \\ &= -(J_0^T J_0)^{-1} [J_0^T r_0 + J_0^T J_0(x^* - x_0)] \\ &= -(J_0^T J_0)^{-1} [J_0^T r^* - J_0^T(r^* - r_0 - J_0(x^* - x_0))]. \end{aligned} \quad (7.2.22)$$

By Theorem 1.2.22, we have

$$\|r^* - r_0 - J_0(x^* - x_0)\| \leq \frac{\gamma}{2} \|x_0 - x^*\|^2. \quad (7.2.23)$$

Noting that $J(x^*)^T r(x^*) = 0$ and using (7.2.17), we get

$$\|J_0^T r^*\| = \|(J_0 - J(x^*))^T r^*\| \leq \sigma \|x - x^*\|. \quad (7.2.24)$$

By using (7.2.20), (7.2.24), (7.2.23) and $\|J_0\| \leq \alpha$, it follows from (7.2.22) that

$$\begin{aligned} \|x_1 - x^*\| &\leq \|(J_0^T J_0)^{-1} (\|J_0^T r^*\| + \|J_0\| \|r^* - r_0 - J_0(x^* - x_0)\|) \\ &\leq \frac{c}{\lambda} \left(\sigma \|x_0 - x^*\| + \frac{\alpha\gamma}{2} \|x_0 - x^*\|^2 \right). \end{aligned} \quad (7.2.25)$$

This proves that (7.2.18) holds at $k = 0$.

Furthermore, from (7.2.25) and (7.2.21), we deduce that

$$\begin{aligned} \|x_1 - x^*\| &\leq \|x_0 - x^*\| \left(\frac{c\sigma}{\lambda} + \frac{c\alpha\gamma}{2\lambda} \|x_0 - x^*\| \right) \\ &\leq \|x_0 - x^*\| \left(\frac{c\sigma}{\lambda} + \frac{\lambda - c\sigma}{2\lambda} \right) \\ &= \frac{c\sigma + \lambda}{2\lambda} \|x_0 - x^*\| \\ &< \|x_0 - x^*\|, \end{aligned} \quad (7.2.26)$$

which shows that (7.2.19) holds at $k = 0$.

For the general case of k , the proof is the same completely as the above. Hence, we complete the proof by induction. $\square$

Theorem 7.2.4 *Assume that the assumptions of Theorem 7.2.2 or Theorem 7.2.3 are satisfied. If $r(x^*) = 0$, then there exists $\epsilon > 0$ such that for any $x_0 \in N(x^*, \epsilon)$, the sequence $\{x_k\}$ generated by Gauss-Newton method converges to x^* with quadratic convergence rate.*

Proof. For Theorem 7.2.2, if $r(x^*) = 0$, then $S(x^*) = 0$. So, the quadratic convergence rate is obtained immediately from (7.2.6).

For Theorem 7.2.3, if $r(x^*) = 0$, then the σ in (7.2.17) can be taken as $\sigma = 0$. Hence, it follows from (7.2.19) that $\{x_k\}$ converges to x^* , and from (7.2.18) that the rate is quadratic. $\square$

Gauss-Newton method now is the most basic method for solving nonlinear least-squares problems. The following example demonstrates that it works well with small residual problems.

Example 7.2.5 Let $r_1(x) = x + 1, r_2(x) = \lambda x^2 + x - 1$. Consider

$$\min f(x) = \sum_{i=1}^2 r_i(x)^2 = (x + 1)^2 + (\lambda x^2 + x - 1)^2,$$

where $n = 1, m = 2$, and $x^* = 0$. For $\lambda = 0.1$, the Gauss-Newton iteration has the following result:

k	1	2	3	4	5	6
x_k	1.000000	0.131148	0.013635	0.001369	0.000137	0.000014

You can see that, when $\lambda = 0.1$, the degree of nonlinearity in $r(x)$ is small, and the Gauss-Newton method works well. In this case, from (7.2.2), the Gauss-Newton iteration is

$$x_{k+1} = \frac{2\lambda^2 x_k^3 + \lambda x_k^2 + 2\lambda x_k}{1 + (2\lambda x_k + 1)^2}.$$

When $\lambda = 0$, in which case $r(x)$ is linear, then $x_1 = 0 = x^*$. This indicates that Gauss-Newton method gets its minimizer in one iteration. When $\lambda \neq 0$, we have

$$x_{k+1} = \lambda x_k + O(\|x_k\|^2).$$

When λ is small enough, the convergence rate is linear. When $|\lambda| > 1$, the Gauss-Newton method fails to converge. This example shows that Gauss-Newton method is valuable only when both x_0 closes to x^* and the matrix $S(x^*)$ is small.

Remark: In practice, we usually use Gauss-Newton method with line search

$$x_{k+1} = x_k - \alpha_k (J(x_k)^T J(x_k))^{-1} J(x_k)^T r(x_k), \quad (7.2.27)$$

which is called the damped Gauss-Newton method, where α_k is a step size. As we have seen, this method guarantees the descent of the objective function in each step and therefore global convergence.

To conclude this section, we mention some numerical aspects on Gauss-Newton method. It should be pointed out that for the problem to solve Gauss-Newton equations

$$J(x_k)^T J(x_k) s = -J(x_k)^T r(x_k), \quad (7.2.28)$$

usually, we employ matrix factorization instead of solving (7.2.28) directly. Then the solution is found by back-substitution technique. So, we can substantially improve the numerical precision. To see this, it follows from the error analysis that

$$\frac{\|\delta s\|}{\|s\|} \leq \kappa(J(x_k)^T J(x_k)) \frac{\|E\|}{\|J(x_k)^T J(x_k)\|}, \quad (7.2.29)$$

where

$$\kappa(J(x_k)^T J(x_k)) = \sigma_1^2 / \sigma_n^2, \quad (7.2.30)$$

δs and E denote the errors of s and $J(x_k)^T J(x_k)$ respectively, and σ_1 and σ_n are the largest and smallest singular values of $J(x_k)$ respectively.

If we employ QR factorization for the augmented matrix, we have

$$\begin{bmatrix} J(x_k) & r_k \end{bmatrix} = Q \begin{bmatrix} R & Q^T r_k \end{bmatrix}, \quad (7.2.31)$$

where Q is an orthogonal matrix,

$$R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix},$$

and R_1 is a nonsingular upper triangular matrix. Then, we obtain

$$J(x_k)^T J(x_k) s = R^T R s = R_1^T R_1 s. \quad (7.2.32)$$

The solution of (7.2.28) can be found by solving

$$R_1 s = -(Q^T r_k)_n, \quad (7.2.33)$$

where $(\cdot)_n$ denotes the first n element partition.

It can be shown that

$$\kappa(R_1) = \frac{\sigma_1}{\sigma_n} \quad (7.2.34)$$

and therefore the errors generated by the orthogonal factorization approach are magnified much less than that directly solve (7.2.28).

7.3 Levenberg-Marquardt Method

7.3.1 Motivation and Properties

Usually, Gauss-Newton method with line search is employed in practice. However, if $J(x)$ is rank-deficient, then either the Gauss-Newton method cannot work well, or the algorithm converges to a non-stationary point.

To overcome the difficulty, we consider employing trust-region technique (for details, see §6.1). In fact, we have seen that, in Gauss-Newton method, a linearized model (7.2.4) is used to replace nonlinear function $r(x)$, and that a linear least-squares problem (7.2.5) is obtained. Unfortunately, this linearization is not effective for all $(x - x_k)$. Therefore, we put a constraint of trust-region on it, and consider the following trust-region model:

$$\min \quad \frac{1}{2} \|J(x_k)(x - x_k) + r(x_k)\|_2^2 \quad (7.3.1)$$

$$\text{s.t.} \quad \|x - x_k\|_2 \leq \Delta_k, \quad (7.3.2)$$

which is a constrained linear least-squares problem. Model (7.3.1)-(7.3.2) can be written as

$$\begin{aligned} \min \quad q_k(x) &= \frac{1}{2} \|r_k\|^2 + r_k^T J(x_k)(x - x_k) + \frac{1}{2} (x - x_k)^T J(x_k)^T J(x_k)(x - x_k) \\ \text{s.t.} \quad &\|x - x_k\|_2 \leq \Delta_k. \end{aligned} \quad (7.3.3)$$

Set $s = x - x_k$. The solution of the subproblem (7.3.1)-(7.3.2) is characterized by solving the system

$$(J(x_k)^T J(x_k) + \mu_k I)s = -J(x_k)^T r(x_k). \quad (7.3.4)$$

Hence,

$$x_{k+1} = x_k - (J(x_k)^T J(x_k) + \mu_k I)^{-1} J(x_k)^T r(x_k). \quad (7.3.5)$$

When $\|(J(x_k)^T J(x_k))^{-1} J(x_k)^T r(x_k)\| \leq \Delta_k$, then $\mu_k = 0$ and the subproblem is solved by s_k . Otherwise, there exists $\mu_k > 0$ such that the solution s_k satisfying $\|s_k\| = \Delta_k$ and

$$(J(x_k)^T J(x_k) + \mu_k I)s_k = -J(x_k)^T r(x_k). \quad (7.3.6)$$

Since $(J(x_k)^T J(x_k) + \mu_k I)$ is positive definite, the direction s produced by (7.3.4) is a descent direction. This method is called the Levenberg-Marquardt method, since it was proposed by Levenberg [199] and Marquardt [210]. The

above discussion exposes that the Levenberg-Marquardt method is just a Gauss-Newton method by replacing the line search with a trust region strategy.

Another perspective about Levenberg-Marquardt method is as follows. This method is just a switch rule between Gauss-Newton method and the steepest descent method. This implies that this method allows choosing any direction between these two directions to be a search direction. When $\mu_k = 0$, it reduces to the Gauss-Newton direction. While μ_k is very large, (7.3.4) is approximate to

$$\mu_k I s = -J(x_k)^T r(x_k). \quad (7.3.7)$$

The produced direction is close to the steepest descent direction.

Furthermore, if, instead of I , we employ some positive definite and diagonal matrix D_k , then (7.3.4) becomes

$$(J(x_k)^T J(x_k) + \mu_k D_k) s = -J(x_k)^T r(x_k). \quad (7.3.8)$$

In this case, the produced direction is a combination of Gauss-Newton direction and the steepest descent direction with respect to a metric matrix D_k .

Next, we will describe some properties of Levenberg-Marquardt method.

Let $s = s(\mu)$ be a solution of

$$(J^T J + \mu I) s = -J^T r, \quad (7.3.9)$$

where, for convenience, $J = J(x)$, $r = r(x)$, and $g = g(x) = J^T r$.

Theorem 7.3.1 *When μ increases monotonically from zero, $\|s(\mu)\|$ will decrease strictly monotonically.*

Proof.

$$\frac{d}{d\mu} \|s\| = \frac{d}{d\mu} (s^T s)^{\frac{1}{2}} = \frac{s^T \frac{ds}{d\mu}}{\|s\|}. \quad (7.3.10)$$

Differentiating (7.3.9) gives

$$(J^T J + \mu I) \frac{ds}{d\mu} = -s. \quad (7.3.11)$$

It follows from (7.3.11) and (7.3.9) that

$$\frac{ds}{d\mu} = (J^T J + \mu I)^{-2} g. \quad (7.3.12)$$

By substituting (7.3.12) into (7.3.10) and using (7.3.9), we obtain

$$\frac{d}{d\mu} \|s\| = -\frac{g^T(J^T J + \mu I)^{-3}g}{\|s\|}. \quad (7.3.13)$$

When $\mu \geq 0$, $J^T J + \mu I$ is positive definite. Therefore (7.3.13) shows that $\|s(\mu)\|$ decreases strictly monotonically. $\square$

Theorem 7.3.2 *The angle ψ between s and $-g$ does not increase monotonically as μ increases.*

Proof. Since

$$\cos \psi = -\frac{g^T s}{\|g\| \|s\|}, \quad (7.3.14)$$

then it is enough to prove

$$\frac{d}{d\mu} \cos \psi \geq 0.$$

By using (7.3.9)-(7.3.13), we deduce

$$\begin{aligned} \frac{d}{d\mu}(\cos \psi) &= \frac{-g^T \frac{ds}{d\mu}}{\|g\| \|s\|} + \frac{g^T s}{\|g\| \|s\|} \frac{\frac{d\|s\|}{d\mu}}{\|s\|} \\ &= \frac{1}{\|g\| \|s\|^3} \{-(g^T(J^T J + \mu I)^{-2}g)^2 \\ &\quad + (g^T(J^T J + \mu I)^{-1}g)(g^T(J^T J + \mu I)^{-3}g)\}. \end{aligned} \quad (7.3.15)$$

So, it is enough to prove that the part in braces is greater than or equal to zero.

Note that $J^T J$ is symmetric, then there is an orthogonal matrix Q such that

$$J^T J = Q^T D Q,$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Set $v = Qg$, then the part in braces on the right-hand-side of (7.3.15) can be written as

$$\sum_{j=1}^n \sum_{k=1}^n \left\{ -\frac{v_j^2 v_k^2}{(\lambda_j + \mu)^2 (\lambda_k + \mu)^2} + \frac{v_j^2 v_k^2}{(\lambda_j + \mu)(\lambda_k + \mu)^3} \right\}. \quad (7.3.16)$$

If let

$$a = \left(\frac{v_1}{(\lambda_1 + \mu)^{3/2}}, \dots, \frac{v_n}{(\lambda_n + \mu)^{3/2}} \right)^T,$$

$$b = \left(\frac{v_1}{(\lambda_1 + \mu)^{1/2}}, \dots, \frac{v_n}{(\lambda_n + \mu)^{1/2}} \right)^T,$$

then (7.3.16) becomes

$$\|a\|^2 \|b\|^2 - \langle a, b \rangle^2,$$

and it follows from Schwartz inequality that the above expression is greater than or equal to zero. Therefore ψ is not increasing. We complete the proof. $\square$

Theorem 7.3.3 *Let $\mu_k > 0$ and s_k be a solution of (7.3.4). Then s_k is a global solution of the subproblem*

$$\min q^{(k)}(s) = \frac{1}{2} \|J_k s + r_k\|_2^2 \quad (7.3.17)$$

$$\text{s.t. } \|s\| \leq \|s_k\|. \quad (7.3.18)$$

Proof. Since s_k is a solution of (7.3.4), then

$$\begin{aligned} q^{(k)}(s_k) &= \frac{1}{2} r_k^T r_k + r_k^T J_k s_k + \frac{1}{2} s_k^T J_k^T J_k s_k \\ &= \frac{1}{2} r_k^T r_k - s_k^T (J_k^T J_k + \mu_k I) s_k + \frac{1}{2} s_k^T J_k^T J_k s_k \\ &= \frac{1}{2} r_k^T r_k - \mu_k s_k^T s_k - \frac{1}{2} s_k^T J_k^T J_k s_k. \end{aligned} \quad (7.3.19)$$

On the other hand, for any s , we have

$$\begin{aligned} q^{(k)}(s) &= \frac{1}{2} r_k^T r_k + s^T J_k^T r_k + \frac{1}{2} s^T J_k^T J_k s \\ &= \frac{1}{2} r_k^T r_k - s^T (J_k^T J_k + \mu_k I) s_k + \frac{1}{2} s^T J_k^T J_k s \\ &= \frac{1}{2} r_k^T r_k - \mu_k s^T s_k - s^T J_k^T J_k s_k + \frac{1}{2} s^T J_k^T J_k s. \end{aligned} \quad (7.3.20)$$

Then, for any s satisfying $\|s\| \leq \|s_k\|$, we deduce that

$$\begin{aligned} q^{(k)}(s) - q^{(k)}(s_k) &= \frac{1}{2} (s_k - s)^T J_k^T J_k (s_k - s) + \mu_k (s_k^T s_k - s^T s_k) \\ &\geq \frac{1}{2} (s_k - s)^T J_k^T J_k (s_k - s) + \mu_k \|s_k\| (\|s_k\| - \|s\|) \\ &\geq 0, \end{aligned} \quad (7.3.21)$$

which shows that s_k is a global optimal solution of problem (7.3.17)-(7.3.18). $\square$

Theorem 7.3.4 *The vector s_k is a solution of problem (7.3.1)-(7.3.2), i.e.,*

$$\min \quad \frac{1}{2} \|J_k s + r_k\|_2^2 \quad (7.3.22)$$

$$\text{s.t.} \quad \|s\| \leq \Delta_k \quad (7.3.23)$$

for some $\Delta_k > 0$ if and only if there exists $\mu \geq 0$ such that

$$(J_k^T J_k + \mu I) s_k = -J_k^T r_k, \quad (7.3.24)$$

$$\mu(\Delta_k - \|s_k\|) = 0, \quad (7.3.25)$$

$$\|s_k\| \leq \Delta_k. \quad (7.3.26)$$

Proof. It is obtained directly from Theorem 6.1.2. $\square$

Usually, Levenberg-Marquardt method is characterized by the equation

$$(J(x_k)^T J(x_k) + \mu_k D(x_k)) s = -J(x_k)^T r(x_k), \quad (7.3.27)$$

where $D(x_k)$ is a diagonal and positive definite matrix. The steplength factor α_k satisfies Armijio rule (2.5.3):

$$f(x_k + \alpha_k s_k) \leq f(x_k) + \sigma \alpha_k g_k^T s_k, \quad \sigma \in \left(0, \frac{1}{2}\right). \quad (7.3.28)$$

Theorem 7.3.5 *For (7.3.27), the condition number of $J(x)^T J(x) + \mu D(x)$ is a non-increasing function of μ .*

Proof. Let β_1 and β_n be the largest and smallest eigenvalues of $D(x)$ respectively. Let λ_1 and λ_n be the largest and smallest eigenvalues of $J(x)^T J(x) + \mu D(x)$ respectively. Let also $\mu_1 > \mu_2 \geq 0$. Since the range of a normal matrix is a convex hull of its spectrum, we have

$$\begin{aligned} \frac{\lambda_1(\mu_1)}{\lambda_n(\mu_1)} &\leq \frac{\lambda_1(\mu_2) + (\mu_1 - \mu_2)\beta_1}{\lambda_n(\mu_2) + (\mu_1 - \mu_2)\beta_n} \\ &\leq \frac{\lambda_1(\mu_2) + (\mu_1 - \mu_2)(1 + \mu_2)^{-1}\lambda_1(\mu_2)}{\lambda_n(\mu_2) + (\mu_1 - \mu_2)(1 + \mu_2)^{-1}\lambda_n(\mu_2)} \\ &= \frac{\lambda_1(\mu_2)}{\lambda_n(\mu_2)}. \end{aligned}$$

Therefore, the conclusion is obtained. $\square$

This property indicates that the Levenberg-Marquardt method improves the condition of the equations solved.

7.3.2 Convergence of Levenberg-Marquardt Method

In this subsection we establish convergence of the Levenberg-Marquardt method.

Theorem 7.3.6 *Let $\{x_k\}$ be a sequence produced by Levenberg-Marquardt method (7.3.27). Suppose that the step lengths α_k are determined by Armijo rule (7.3.28). If there is a subsequence $\{x_{k_i}\}$ that converges to x^* , and if the corresponding subsequence $\{J_{k_i}^T J_{k_i} + \mu_{k_i} D_{k_i}\}$ converges to some positive definite matrix P , where $J_{k_i} = J(x_{k_i})$ and $D_{k_i} = D(x_{k_i})$ denoting a diagonal positive definite matrix, then $g(x^*) = 0$.*

Proof. (By contradiction) Suppose that $g(x^*) \neq 0$. Let

$$\begin{aligned} s_{k_i} &= -(J_{k_i}^T J_{k_i} + \mu_{k_i} D_{k_i})^{-1} J_{k_i}^T r_{k_i}, \\ s^* &= \lim s_{k_i} = -P^{-1} J(x^*)^T r(x^*), \end{aligned}$$

where $r_{k_i} = r(x_{k_i})$. Obviously, $g(x^*)^T s^* < 0$. Let $\beta \in (0, 1)$, $\sigma \in (0, \frac{1}{2})$. Let also m^* be the least non-negative integer m such that

$$f(x^* + \beta^m s^*) < f(x^*) + \sigma \beta^m g(x^*)^T s(x^*).$$

By continuity, for k sufficiently large, we have

$$f(x_{k_i} + \beta^{m^*} s_{k_i}) \leq f(x_{k_i}) + \sigma \beta^{m^*} g(x_{k_i})^T s_{k_i}.$$

Hence

$$f(x_{k_i+1}) = f(x_{k_i} + \beta^{m_{k_i}} s_{k_i}) \leq f(x_{k_i}) + \sigma \beta^{m^*} g(x_{k_i})^T s_{k_i}. \quad (7.3.29)$$

By the monotone descent of the method, we have

$$\lim f(x_{k_i+1}) = \lim f(x_{k_i}) = f(x^*).$$

Therefore, taking limits on both sides of (7.3.29) yields

$$f(x^*) \leq f(x^*) + \sigma \beta^{m^*} g(x^*)^T s^* < 0.$$

This is impossible because $\sigma \beta^{m^*} g(x^*)^T s^* < 0$. So we complete the proof. $\square$

The above theorem states the convergence of a subsequence. Below, we give convergence of the whole sequence.

Theorem 7.3.7 *Suppose that the following assumptions hold:*

(a) *the level set*

$$L(\bar{x}) = \{x \mid f(x) \leq f(\bar{x})\}$$

is bounded and closed for any $\bar{x} \in R^n$;

(b) *the number of stationary points at which the function values of $f(x)$ are the same is finite;*

(c) *$J(x)^T J(x)$ is positive definite $\forall x$;*

(d) *$\mu_k \leq M < \infty, \forall k$, that is, M is an upper bound of μ_k .*

Then for any initial point x_0 , the sequence $\{x_k\}$ generated from Levenberg-Marquardt method converges to a stationary point of $f(x)$.

Proof. From (a) and the monotone property of iterative function, we know that the sequence $\{x_k\}$ is in compact set $L(\bar{x})$. This shows that $\{x_k\}$ must have accumulation points. To prove the theorem, we only need to prove the accumulation points are unique.

By (c), (d) and Theorem 7.3.6, we have that each accumulation point of $\{x_k\}$ is unique. Since $\{f(x)\}$ is a monotone descent sequence, $f(x)$ has the same values at accumulation points of $\{x_k\}$. Also, from (b), it follows that the number of stationary points of f on $L(\bar{x})$ are finite. Therefore, there are only finitely many accumulation points.

Notice that, for some subsequence $\{x_{k_i}\}$, we have $x_{k_i} \rightarrow \hat{x}_k$ and

$$\lim_{k \rightarrow \infty} g(x_{k_i}) = g(\hat{x}_k) = 0.$$

Notice also that

$$s(\mu_{k_i}) = -(J(x_{k_i})^T J(x_{k_i}) + \mu_{k_i} D(x_{k_i}))^{-1} g(x_{k_i}).$$

Then it follows from (c) and (d) that $s(\mu_{k_i}) \rightarrow 0$. Therefore, for sequence $\{s(\mu_k)\}$, we have $s(\mu_k) \rightarrow 0$.

Assume for the moment that there are more than one accumulation point of $\{x_k\}$. Let ϵ^* be the smallest distance between any two accumulation points. Since $\{x_k\}$ is in a compact set, there exists a positive integer N , such that for all $k \geq N$, x_k is contained in a closed ball with some accumulation point

as center and $\epsilon^*/4$ as radius. On the other hand, there is an integer $N' \geq N$ such that

$$\|s(\mu_k)\| < \epsilon^*/4, \quad \forall k \geq N'.$$

Therefore, when $k \geq N'$, all x_k are in the closed ball mentioned above with that accumulation point as center and $\epsilon^*/4$ as radius. Then we have a contradiction which proves the theorem. $\square$

The above theorem establishes global convergence of the Levenberg-Marquardt method. In the following, similar to Theorem 7.2.2, we discuss the convergence rate of the Levenberg-Marquardt method.

Theorem 7.3.8 *Suppose that the iterates x_k generated by Levenberg-Marquardt method converge to a stationary point x^* . Let l be the smallest eigenvalue of $J(x^*)^T J(x^*)$, M the maximum of absolute values of eigenvalues of $S(x^*) = \sum_{i=1}^m r_i(x^*) \nabla^2 r_i(x^*)$. If*

$$\tau = M/l < 1, \quad 0 < \beta < (1 - \tau)/2, \quad \mu_k \rightarrow 0, \quad (7.3.30)$$

then, for all k sufficiently large, the stepsize $\alpha_k = 1$,

$$\limsup \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq \tau, \quad (7.3.31)$$

and x^ is a strict local minimizer of $f(x)$.*

Proof. We first prove $\alpha_k = 1$ for k large enough. Notice that

$$f(x_k + s_k) - f(x_k) = g_k^T s_k + \frac{1}{2} s_k^T G(x_k + \theta s_k) s_k, \quad (7.3.32)$$

where $\theta \in (0, 1)$. By means of Armijo rule (7.3.28), to prove $\alpha_k = 1$ for k large enough, we need to show

$$\beta g_k^T s_k - [f(x_k + s_k) - f(x_k)] \geq 0. \quad (7.3.33)$$

By use of $g_k = -(J_k^T J_k + \mu_k D_k) s_k$ and (7.3.32), the left-hand side of (7.3.33) can be written as

$$\begin{aligned} & (1 - \beta) s_k^T (J_k^T J_k + \mu_k D_k) s_k - \frac{1}{2} s_k^T G(x_k + \theta s_k) s_k \\ &= s_k^T \left[(1 - \beta) J_k^T J_k - \frac{1}{2} G(x_k) + (1 - \beta) \mu_k D_k \right. \\ & \quad \left. - \frac{1}{2} (G(x_k + \theta s_k) - G(x_k)) \right] s_k \\ &= s_k^T \left[\left(\frac{1}{2} - \beta \right) J_k^T J_k - \frac{1}{2} S(x_k) + V_k \right] s_k, \end{aligned}$$

where $V_k = (1 - \beta)\mu_k D_k - \frac{1}{2}(G(x_k + \theta s_k) - G(x_k))$, $S(x_k)$ is defined by (7.1.7). Since $V_k \rightarrow 0$, to prove (7.3.33) holds for k large enough, we show $(\frac{1}{2} - \beta)J_k^T J_k - \frac{1}{2}S(x_k)$ converges to a positive definite matrix. Note that the smallest eigenvalue of

$$\left(\frac{1}{2} - \beta\right) J(x^*)^T J(x^*) - \frac{1}{2}S(x^*)$$

is bounded below and that the lower bound is

$$\left(\frac{1}{2} - \beta\right) l - \frac{1}{2}M = l \left[\frac{1}{2} - \beta - \frac{1}{2}\tau \right] > 0,$$

which holds because the second inequality in (7.3.30) is met for β . So we obtain $\alpha_k = 1$ for sufficiently large k .

Second, we prove (7.3.31). By (7.3.27) and (7.1.7), we have

$$\begin{aligned} x_{k+1} - x^* &= x_k - x^* - (J_k^T J_k + \mu_k D_k)^{-1} g_k \\ &= x_k - x^* - (J_k^T J_k + \mu_k D_k)^{-1} [G_k(x_k - x^*) \\ &\quad + g_k + G_k(x^* - x_k)] \\ &= -(J_k^T J_k + \mu_k D_k)^{-1} [S(x_k)(x_k - x^*) \\ &\quad - \mu_k D_k(x_k - x^*) + g_k + G_k(x^* - x_k)]. \end{aligned} \quad (7.3.34)$$

Taking norm gives

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \|(J_k^T J_k)^{-1}(\|S(x_k)\| \|x_k - x^*\| \\ &\quad + \mu_k \|D_k\| \|x_k - x^*\| + \|g_k + G_k(x^* - x_k)\|)\|. \end{aligned} \quad (7.3.35)$$

Since

$$\begin{aligned} \|g_k + G_k(x^* - x_k)\| &= \|g_k - g(x^*) - G_k(x_k - x^*)\| \\ &\leq \epsilon_k \|x_k - x^*\|, \end{aligned} \quad (7.3.36)$$

where $\epsilon_k \rightarrow 0$, then dividing the both sides of (7.3.35) by $\|x_k - x^*\|$ deduces

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq \|(J_k^T J_k)^{-1}(\|S(x_k)\| + \mu_k \|D_k\| + \epsilon_k)\|. \quad (7.3.37)$$

Note that $\mu_k \rightarrow 0$ and that $\epsilon_k \rightarrow 0$, and it follows immediately that

$$\limsup \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq \frac{M}{l} = \tau,$$

which proves (7.3.31).

Finally, since $g(x^*) = 0$ and $G(x^*) = J(x^*)^T J(x^*) + S(x^*)$ with the lower bound $l - M > 0$ of the smallest eigenvalue, then $G(x^*)$ is positive definite. Hence x^* is a strict local minimizer of $f(x)$. $\square$

As mentioned above, the Levenberg-Marquardt method can be described and analyzed by use of the framework of trust region method (7.3.1)–(7.3.2) or (7.3.3). So, following the discussions of Section 6.1, we immediately have the following algorithm and theorem which are straightforward consequences of Algorithm 6.1.1 and Theorem 6.1.9, respectively.

Algorithm 7.3.9 (*Trust-Region Type Levenberg-Marquardt Algorithm*)

Step 1. Given initial point $x_0, \bar{\Delta}, \Delta_0 \in (0, \bar{\Delta}), \epsilon \geq 0, 0 < \eta_1 \leq \eta_2 < 1$ and $0 < \gamma_1 < 1 < \gamma_2, k := 0$.

Step 2. If $\|g_k\| = \|J_k^T r_k\| \leq \epsilon$, stop.

Step 3. Approximately solve the subproblem (7.3.1)–(7.3.2) for s_k .

Step 4. Compute

$$\begin{aligned} \text{Pred}_k &= f(x_k) - q_k(s_k), \\ \text{Ared}_k &= f(x_k) - f(x_k + s_k), \\ r_k &= \frac{\text{Ared}_k}{\text{Pred}_k}. \end{aligned}$$

Step 5. If $r_k < \eta_1$, set $\Delta_k = \gamma_1 \Delta_k$ and go to Step 3.

Step 6. Set $x_{k+1} = x_k + s_k$. Set

$$\Delta_{k+1} = \begin{cases} \min\{\gamma_2 \Delta_k, \bar{\Delta}\}, & \text{if } r_k \geq \eta_2 \text{ and } \|s_k\| = \Delta_k, \\ \Delta_k, & \text{otherwise.} \end{cases}$$

Step 7. Set $k := k + 1$, go to Step 2. $\square$

From Step 3 of the above algorithm, s_k is the approximate solution of subproblem (7.3.1)–(7.3.2). It follows from Lemma 6.1.3 that

$$q_k(0) - q_k(s_k) \geq c_1 \|J_k^T r_k\| \min \left(\Delta_k, \frac{\|J_k^T r_k\|}{\|J_k^T J_k\|} \right) \quad (7.3.38)$$

for some constant $c_1 > 0$.

Now we can state the convergence result which is a straightforward consequence of Theorem 6.1.9.

Theorem 7.3.10 *Suppose that the function $f(x) = \frac{1}{2} \sum_{i=1}^m [r_i(x)]^2$ is twice continuously differentiable, that the level set*

$$\mathcal{L}(x_0) = \{x \mid f(x) \leq f(x_0)\}$$

is bounded, and that there are constants $M_1 > 0, M_2 > 0$ such that

$$\begin{aligned} \|\nabla^2 f(x)\| &\leq M_1, \quad \forall x \in \mathcal{L}(x_0), \\ \|J(x)^T J(x)\| &\leq M_2, \quad \forall x \in \mathcal{L}(x_0). \end{aligned}$$

Then we have that

$$\lim_{k \rightarrow \infty} \nabla f(x_k) = \lim_{k \rightarrow \infty} J_k^T r_k = 0. \quad (7.3.39)$$

7.4 Implementation of L-M Method

There are various implementations of the Levenberg-Marquardt method. Moré [218] gave an efficient and reliable implementation, which is contained in the MINPACK software package.

The Levenberg-Marquardt method Moré [218] considered is to find s by means of solving equations

$$s(\mu) = -(J_k^T J_k + \mu_k D_k^T D_k)^{-1} J_k^T r_k, \quad (7.4.1)$$

which correspond to a trust region subproblem (or a constrained linear least-squares problem)

$$\min \|r_k + J_k s\| \quad \text{s.t.} \quad \|D_k s\| \leq \Delta_k, \quad (7.4.2)$$

where $\Delta_k > 0$ is the trust-region radius. If J_k is singular and $\mu_k = 0$, the solution of (7.4.2) can be defined by a limit

$$D_k s(0) = \lim_{\mu_k \rightarrow 0^+} D_k s(\mu_k) = -(J_k D_k^{-1})^+ r_k. \quad (7.4.3)$$

There are two possibilities: either $\mu_k = 0$ and $\|D_k s(0)\| \leq \Delta_k$, in which case $s(0)$ is the solution of (7.4.2); or $\mu_k > 0$ and $\|D_k s(\mu_k)\| = \Delta_k$, in which case $s(\mu_k)$ is a unique solution of (7.4.2). Hence we can describe the following algorithm.

Algorithm 7.4.1 (*Levenberg-Marquardt Algorithm*)

(a) Given $\Delta_k > 0$, find $\mu_k \geq 0$ such that

$$(J_k^T J_k + \mu_k D_k) s_k = -J_k^T r_k.$$

Then either $\mu_k = 0$ and $\|D_k s_k\| \leq \Delta_k$, or $\mu_k > 0$ and $\|D_k s_k\| = \Delta_k$.

(b) If $\|r(x_k + s_k)\| \leq \|r(x_k)\|$, set $x_{k+1} = x_k + s_k$ and compute J_{k+1} ; otherwise set $x_{k+1} = x_k$ and $J_{k+1} = J_k$.

(c) Choose Δ_{k+1} and D_{k+1} . $\square$

In the following, we discuss how to perform the above algorithm efficiently and reliably.

(1) How to solve the trust-region subproblem (i.e., constrained linear least-squares problem).

For equations

$$(J_k^T J_k + \mu_k D_k^T D_k) s = -J_k^T r_k, \quad (7.4.4)$$

the simplest way is using Cholesky factorization. However, because of the special structure of the coefficient matrix in (7.4.4), it is easy to use QR factorization.

Note that (7.4.4) are just the normal equations for linear least-squares problem

$$\begin{bmatrix} J_k \\ \mu_k^{1/2} D_k \end{bmatrix} s \cong - \begin{bmatrix} r \\ 0 \end{bmatrix}. \quad (7.4.5)$$

For the structure of (7.4.5), instead of computing $J_k^T J_k$ and $D_k^T D_k$ and their Cholesky factorization, we can use column pivoting QR factorization.

Now we describe the two-step QR factorization to find the solution of the linear least-squares problem (7.4.5).

First Step: Calculate the QR factorization of J_k and obtain

$$Q J_k \pi = \begin{bmatrix} T & W \\ 0 & 0 \end{bmatrix}, \quad (7.4.6)$$

where Q is orthogonal, T is a nonsingular upper triangular matrix with $\text{rank}(T) = \text{rank}(J_k)$, and π is a permutation matrix. If $\mu_k = 0$, then the

solution of (7.4.5) is

$$s = \pi \begin{bmatrix} T^{-1} & 0 \\ 0 & 0 \end{bmatrix} Q r_k \equiv J_k^- r_k, \quad (7.4.7)$$

where J_k^- denotes $\{1, 3\}$ -inverse satisfying

$$J_k J_k^- J_k = J_k, \quad J_k J_k^- = (J_k J_k^-)^T$$

(see X.He and W. Sun [172], Ben-Israel and Greville [12]). If $\mu_k > 0$, since (7.4.6) becomes

$$\begin{bmatrix} Q & 0 \\ 0 & \pi^T \end{bmatrix} \begin{bmatrix} J_k \\ \mu_k^{1/2} D_k \end{bmatrix} \pi = \begin{bmatrix} R \\ 0 \\ D_\mu \end{bmatrix}, \quad (7.4.8)$$

where $D_\mu = \mu_k^{1/2} \pi^T D_k \pi$ and R is an upper trapezoid matrix, it follows from (7.4.8) that (7.4.5) becomes

$$\begin{bmatrix} R \\ 0 \\ D_\mu \end{bmatrix} \pi^T s = - \begin{bmatrix} Q r \\ 0 \end{bmatrix}. \quad (7.4.9)$$

Second Step: It is easy to eliminate D_μ in (7.4.9) by a sequence of $n(n+1)/2$ Givens rotations and obtain

$$W \begin{bmatrix} R \\ 0 \\ D_\mu \end{bmatrix} = \begin{bmatrix} R_\mu \\ 0 \\ 0 \end{bmatrix}, \quad (7.4.10)$$

where R_μ is a nonsingular upper triangular matrix and W a product of a sequence of rotations. Then (7.4.9) becomes

$$\begin{bmatrix} R_\mu \\ 0 \end{bmatrix} \pi^T s = -W \begin{bmatrix} Q r \\ 0 \end{bmatrix} \triangleq \begin{bmatrix} u \\ v \end{bmatrix}, \quad (7.4.11)$$

and we obtain

$$s = \pi R_\mu^{-1} u. \quad (7.4.12)$$

(2) How to update the trust-region radius Δ_k .

As described in §6.1, the choice of Δ_k depends on the ratio between actual reduction and predicted reduction of the objective function. In the nonlinear least-squares case, the ratio is

$$\rho = \frac{\|r(x_k)\|^2 - \|r(x_k + s_k)\|^2}{\|r(x_k)\|^2 - \|r(x_k) + J(x_k)s_k\|^2}, \quad (7.4.13)$$

which measures the agreement between the linearized model and the nonlinear function. For example, if $r(x)$ is linear, then $\rho = 1$. If $J(x_k)^T r(x_k) \neq 0$, then $\rho \rightarrow 1$ when $\|s_k\| \rightarrow 0$. If $\|r(x_k + s_k)\| \geq \|r(x_k)\|$, then $\rho \leq 0$.

Because of roundoff error, there may be overflow when we compute ρ by (7.4.13). So we write (7.4.13) in a safeguard form. Multiplying both sides of (7.4.4) by $2s^T$ yields

$$-2r_k^T J_k^T s = 2s^T J_k^T J_k s + 2\mu_k s D_k^T D_k s,$$

which is

$$r_k^T r_k - r_k^T r_k - 2r_k^T J_k^T s - s^T J_k^T J_k s = s^T J_k^T J_k s + 2\mu_k s^T D_k^T D_k s.$$

Then we obtain

$$\|r_k\|^2 - \|r_k + J_k s\|^2 = \|J_k s\|^2 + 2\mu_k \|D_k s\|^2. \quad (7.4.14)$$

Substituting the above into (7.4.13) gives

$$\rho = \frac{1 - \left[\frac{\|r(x_k + s_k)\|}{\|r(x_k)\|} \right]^2}{\left[\frac{\|J_k s\|}{\|r(x_k)\|} \right]^2 + 2 \left[\frac{\mu_k^{1/2} \|D_k s\|}{\|r(x_k)\|} \right]^2}. \quad (7.4.15)$$

It is easy to see from (7.4.14) that

$$\|J_k s\| \leq \|r(x_k)\|, \quad \mu_k^{1/2} \|D_k s\| \leq \|r(x_k)\|.$$

Hence the computation in (7.4.15) will not lead to overflow. Also, regardless of roundoff error, the denominator in (7.4.15) is always nonnegative. It should be mentioned that when $\|r(x_k + s_k)\| \gg \|r(x_k)\|$, the numerator in (7.4.15) may be overflowed. However, since we are only interested in $\rho \geq 0$, then when $\|r(x_k + s_k)\| > \|r(x_k)\|$, we set $\rho = 0$ without needing to compute ρ by (7.4.15).

(3) How to find a Levenberg-Marquardt parameter.

In the Moré algorithm, if

$$|\phi(\mu)| \leq \sigma \Delta, \quad \sigma \in (0, 1), \quad (7.4.16)$$

where

$$\phi(\mu) = \|D(J^T J + \mu D^T D)^{-1} J^T r\| - \Delta, \quad (7.4.17)$$

then $\mu > 0$ is accepted as a Levenberg-Marquardt parameter, where σ indicates the related error in $\|D_k s(\mu)\|$. If $\phi(0) \leq 0$, then $\mu = 0$ is a required parameter. Therefore we only need to discuss the case of $\phi(0) > 0$. Since ϕ is a continuous and strictly descending function, then, when $\mu \rightarrow \infty$, we have $\phi(\mu) \rightarrow -\Delta$. Thus, there exists a unique $\mu^* > 0$ such that $\phi(\mu^*) = 0$. To determine the Levenberg-Marquardt parameter, we start from $\mu_0 > 0$ and generate a sequence $\{\mu_k\} \rightarrow \mu^*$.

From (7.4.17), we have

$$\phi(\mu) = \|(\tilde{J}^T \tilde{J} + \mu I)^{-1} \tilde{J}^T r\| - \Delta, \quad (7.4.18)$$

where $\tilde{J} = JD^{-1}$. Let $\tilde{J} = U\Sigma V^T$ be the singular value decomposition of $\tilde{J}$, then

$$\phi(\mu) = \sum_{i=1}^n \left[\frac{\sigma_i^2 z_i^2}{(\sigma_i^2 + \mu)^2} \right]^{1/2} - \Delta, \quad (7.4.19)$$

where $z = U^T r$, $\sigma_1, \dots, \sigma_n$ are singular values of $\tilde{J}$. Hence we assume

$$\phi(\mu) \doteq \frac{a}{b + \mu} \equiv \tilde{\phi}(\mu) \quad (7.4.20)$$

and choose a and b such that $\tilde{\phi}(\mu_k) = \phi(\mu_k)$, $\tilde{\phi}'(\mu_k) = \phi'(\mu_k)$. Then $\tilde{\phi}(\mu_{k+1}) = 0$ if

$$\mu_{k+1} = \mu_k - \left[\frac{\phi(\mu_k) + \Delta}{\Delta} \right] \left[\frac{\phi(\mu_k)}{\phi'(\mu_k)} \right]. \quad (7.4.21)$$

To make computation of μ_{k+1} safe and reliable, the Moré algorithm designs the following technique for computing μ_{k+1} .

Let

$$u_0 = \frac{\|(JD^{-1})^T r\|}{\Delta},$$

$$l_0 = \begin{cases} -\phi(0)/\phi'(0), & \text{if } J \text{ is nonsingular,} \\ 0, & \text{otherwise,} \end{cases}$$

- (a) If $\mu_k \notin (l_k, u_k)$, set $\mu_k = \max\{0.001u_k, (l_k u_k)^{1/2}\}$.
 (b) Compute $\phi(\mu_k)$ and $\phi'(\mu_k)$. Update u_k :

$$u_{k+1} = \begin{cases} \mu_k, & \text{if } \phi(\mu_k) < 0, \\ u_k, & \text{otherwise.} \end{cases}$$

Update l_k :

$$l_{k+1} = \max \left\{ l_k, \mu_k - \frac{\phi(\mu_k)}{\phi'(\mu_k)} \right\}.$$

- (c) Compute μ_{k+1} by (7.4.21).

The above algorithm gives upper and lower bounds of μ_k . In (a), it shows that if μ_k is not in (l_k, u_k) , a point in (l_k, u_k) inclining to l_k will replace μ_k . In (b), the convexity of ϕ guarantees that the Newton's iteration can be used to update l_k . The sequence $\{\mu_k\}$ generated by the algorithm will converge to μ^* . In fact, if we set $\sigma = 0.1$, it takes no more than two steps on average to satisfy (7.4.16).

As to computing $\phi'(\mu)$, we have from (7.4.17) that

$$\phi'(\mu) = -\frac{(D^T q(\mu))^T (J^T J + \mu D^T D)^{-1} (D^T q(\mu))}{\|q(\mu)\|}, \quad (7.4.22)$$

where $q(\mu) = Ds(\mu)$. By (7.4.8) and (7.4.10) we get

$$\pi^T (J^T J + \mu D^T D) \pi = R_\mu^T R_\mu.$$

Then

$$\phi'(\mu) = -\|q(\mu)\| \left\| R_\mu^{-T} \left(\frac{\pi^T D^T q(\mu)}{\|q(\mu)\|} \right) \right\|^2. \quad (7.4.23)$$

(4) How to update the scaling matrix.

In the Levenberg-Marquardt method, D_k is a diagonal matrix which reduces the effects of poor scaling of the problems. In the algorithm, we choose

$$D_k = \text{diag}(d_1^{(k)}, \dots, d_n^{(k)}), \quad (7.4.24)$$

where

$$\begin{aligned} d_i^{(0)} &= \|\partial_i r(x_0)\|, \\ d_i^{(k)} &= \max\{d_i^{(k-1)}, \|\partial_i r(x_k)\|\}, \quad k \geq 1. \end{aligned}$$

We should point out that the above scaling is invariant under scaling, that is, if D is a diagonal and positive definite matrix, then for function $r(x)$ with starting point x_0 and for function $\tilde{r}(x) = r(D^{-1}x)$ with starting point $\tilde{x}_0 = Dx_0$, Algorithm 7.4.1 generates the same sequence of iterates.

Finally, we give the Moré version of the Levenberg-Marquardt algorithm and the convergence theorem.

Algorithm 7.4.2 (*Moré's Version*)

- (a) Let $\sigma \in (0, 1)$. If $\|D_k J_k^- r_k\| \leq (1 + \sigma)\Delta_k$, set $\mu_k = 0$ and $s_k = -J_k^- r_k$; otherwise, determine $\mu_k > 0$ such that if

$$\begin{bmatrix} J_k \\ \mu_k^{1/2} D_k \end{bmatrix} s_k \cong - \begin{bmatrix} r_k \\ 0 \end{bmatrix}$$

(i.e., s_k is the solution of the above least-squares problem), then

$$(1 - \sigma)\Delta_k \leq \|D_k s_k\| \leq (1 + \sigma)\Delta_k.$$

- (b) Compute the ratio ρ_k between the actual reduction and the predicted reduction of the objective function.
- (c) If $\rho_k \leq 0.0001$, set $x_{k+1} = x_k$ and $J_{k+1} = J_k$.
If $\rho_k > 0.0001$, set $x_{k+1} = x_k + s_k$, and compute J_{k+1} .
- (d) If $\rho_k \leq \frac{1}{4}$, set $\Delta_{k+1} \in [\frac{1}{10}\Delta_k, \frac{1}{2}\Delta_k]$. If either $\rho_k \in [\frac{1}{4}, \frac{1}{3}]$ and $\mu_k = 0$, or $\rho_k \geq \frac{3}{4}$, then set $\Delta_{k+1} = 2\|D_k s_k\|$.
- (e) Update D_k by (7.4.24). $\square$

For the above algorithm, the convergence theorem is stated as follows without proof. The interested readers may consult Moré [218].

Theorem 7.4.3 Let $r : R^n \rightarrow R^m$ be continuously differentiable. Let $\{x_k\}$ be a sequence generated by Algorithm 7.4.2. Then

$$\liminf_{k \rightarrow +\infty} \|(J_k D_k^{-1})^T r_k\| = 0. \quad (7.4.25)$$

This result indicates that the scaled gradient is, at last, sufficiently small. If $\{J_k\}$ is bounded, then (7.4.25) implies

$$\liminf_{k \rightarrow +\infty} \|J_k^T r_k\| = 0. \quad (7.4.26)$$

Further, if $\nabla r(x)$ is uniformly continuous, then

$$\lim_{k \rightarrow +\infty} \|J_k^T r_k\| = 0. \quad (7.4.27)$$

7.5 Quasi-Newton Method

We have seen from the above sections that, for large-residual problems (i.e., $r(x)$ is large or $r(x)$ is severely nonlinear), the performance of the Gauss-Newton method and Levenberg-Marquardt method is usually poor. The convergence is slow and only linear. This is because we don't use the second-order information $S(x)$ in Hessian $G(x) = J(x)^T J(x) + S(x)$ which is significant. As mentioned in §7.1, in fact, computation of $S(x)$ is either difficult or expensive. It is also not suitable to use the secant approximation of the whole Hessian $G(x)$. So, it may be a good idea to use a secant approximation of the second information $S(x) = \sum_{i=1}^m r_i(x) \nabla^2 r_i(x)$ in $G(x)$.

Let B_k be a secant approximation of $S(x_k)$, then the iteration (7.1.10) becomes

$$(J(x_k)^T J(x_k) + B_k) d_k = -J(x_k)^T r(x_k). \quad (7.5.1)$$

Since

$$S(x_{k+1}) = \sum_{i=1}^m r_i(x_{k+1}) \nabla^2 r_i(x_{k+1}), \quad (7.5.2)$$

then we use

$$B_{k+1} = \sum_{i=1}^m r_i(x_{k+1}) (H_i)_{k+1} \quad (7.5.3)$$

to approximate $S(x_{k+1})$, where $(H_i)_{k+1}$ is a secant approximation of $\nabla^2 r_i(x_{k+1})$. Then we have that

$$(H_i)_{k+1} (x_{k+1} - x_k) = \nabla r_i(x_{k+1}) - \nabla r_i(x_k). \quad (7.5.4)$$

Hence, we get

$$\begin{aligned} B_{k+1} (x_{k+1} - x_k) &= \sum_{i=1}^m r_i(x_{k+1}) (H_i)_{k+1} (x_{k+1} - x_k) \\ &= \sum_{i=1}^m r_i(x_{k+1}) (\nabla r_i(x_{k+1}) - \nabla r_i(x_k)) \\ &= (J(x_{k+1}) - J(x_k))^T r(x_{k+1}) \triangleq y_k \end{aligned} \quad (7.5.5)$$

which is a quasi-Newton condition imposed on B_k .

Similarly, if we ask

$$(J(x_{k+1})^T J(x_{k+1}) + B_{k+1})s_k = J(x_{k+1})^T r(x_{k+1}) - J(x_k)^T r(x_k) \quad (7.5.6)$$

to hold, then B_{k+1} should satisfy

$$B_{k+1}s_k = \tilde{y}_k, \quad (7.5.7)$$

where

$$\tilde{y}_k = J(x_{k+1})^T r(x_{k+1}) - J(x_k)^T r(x_k) - J(x_{k+1})^T J(x_{k+1})s_k. \quad (7.5.8)$$

Now, we give an update formula for B_k by weighted Frobenius norm. The following theorem is a restatement of Theorem 5.1.10 in Chapter 5.

Theorem 7.5.1 *Let $v_k^T s_k > 0$ and $T \in R^{n \times n}$ be a symmetric and positive definite matrix such that*

$$TT^T s_k = v_k, \quad (7.5.9)$$

where

$$\begin{aligned} v_k &\triangleq \nabla f(x_{k+1}) - \nabla f(x_k) \\ &= J(x_{k+1})^T r(x_{k+1}) - J(x_k)^T r(x_k). \end{aligned} \quad (7.5.10)$$

Then the update

$$\begin{aligned} B_{k+1} &= B_k + \frac{(y_k - B_k s_k)v_k^T + v_k(y_k - B_k s_k)^T}{s_k^T v_k} \\ &\quad - \frac{s_k^T (y_k - B_k s_k)}{(s_k^T v_k)^2} v_k v_k^T \end{aligned} \quad (7.5.11)$$

is a unique solution of the minimization problem

$$\begin{aligned} \min \quad & \|T^{-T}(B_{k+1} - B_k)T^{-1}\|_F \\ \text{s.t.} \quad & (B_{k+1} - B_k) \text{ is symmetric, } B_{k+1}s_k = y_k. \end{aligned} \quad (7.5.12)$$

Dennis, Gay and Welsch [88] use the quasi-Newton condition (7.5.5) and (7.5.11), and present a quasi-Newton algorithm NL2SOL with trust region strategy. At each step, it is required to solve the subproblem

$$\begin{aligned} \min \quad & \frac{1}{2}r(x_k)^T r(x_k) + (x - x_k)^T J(x_k)^T r(x_k) \\ & + \frac{1}{2}(x - x_k)^T (J(x_k)^T J(x_k) + B_k)(x - x_k) \\ \text{s.t.} \quad & \|x - x_k\| \leq \Delta_k. \end{aligned} \quad (7.5.13)$$

In this algorithm, a deficiency of the update (7.5.11) for B_k is that this matrix is not guaranteed to vanish when the iterates approach to a zero-residual solution, so it can interfere with superlinear convergence. This problem can be avoided by a strategy of scaling B_k , that is, we choose a scaling factor

$$\gamma_k = \min \left\{ \frac{s_k^T y_k}{s_k^T B_k s_k}, 1 \right\}, \quad (7.5.14)$$

multiply B_k by γ_k , and then use (7.5.11) to update it.

Numerical experiments show that, for large-residual problems, quasi-Newton algorithm NL2SOL is significantly advantageous; for small-residual problems, the performance of NL2SOL and Moré's Levenberg-Marquardt algorithm is similar; for zero-residual problems we prefer the Gauss-Newton method. Therefore, the Gauss-Newton method, Levenberg-Marquardt method, and quasi-Newton method introduced in this chapter are very important to solve nonlinear least-squares problems. Now, Moré's Levenberg-Marquardt algorithm and quasi-Newton algorithm NL2SOL are very popular.

Similar to the above discussion, Bartholomew-Biggs [15] uses the quasi-Newton condition (7.5.5) and the following rank-one updating formula

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}, \quad (7.5.15)$$

and gives a quasi-Newton method for nonlinear least-squares problems. In Bartholomew-Biggs' algorithm, the scaling factor is

$$\gamma_k = r_{k+1}^T r_{k+1} / r_k^T r_k. \quad (7.5.16)$$

Fletcher and Xu [139] presented a hybrid algorithm which combines Gauss-Newton method and quasi-Newton method. If the current Gauss-Newton step reduces the function f by a certain fixed amount, i.e.,

$$f(x_k) - f(x_{k+1}) \geq \tau f(x_k), \quad \tau \in (0, 1), \quad (7.5.17)$$

we use the Gauss-Newton step. Otherwise, we use quasi-Newton update (for example, BFGS update). In the zero-residual case, the method eventually takes Gauss-Newton steps and gives quadratic convergence; while in the nonzero-residual case, the method eventually reduces to BFGS formula. The theoretical analysis shows that Fletcher-Xu method is superlinearly convergent. Normally, we take $\tau = 0.2$ in (7.5.17).

Exercises

1. Solve the least-squares problem

$$\min f(x) = \frac{1}{2}[(x_2 - x_1^2)^2 + (1 - x_1)^2], \quad x^{(0)} = (0, 0)^T$$

by Gauss-Newton method and Levenberg-Marquardt method.

2. Consider nonlinear least-squares problems:

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} r(x)^T r(x) = \frac{1}{2} \sum_{i=1}^m [r_i(x)]^2, \quad m \geq n$$

where

$$\begin{aligned} r_1(x) &= x_1^2 + x_2^2 + x_3^2 - 1, \\ r_2(x) &= x_1^2 + x_2^2 + (x_3 - 2)^2 - 1, \\ r_3(x) &= x_1 + x_2 + x_3 - 1, \\ r_4(x) &= x_1 + x_2 - x_3 + 1, \\ r_5(x) &= x_1^3 + 3x_2^2 + (5x_3 - x_1 + 1)^2 - 36. \end{aligned}$$

- (1) Compute $\nabla f(x)$, $J(x)^T J(x)$, and $\nabla^2 f(x)$.
 (2) Please answer whether $J(x)^T J(x) = \nabla^2 f(x)$ holds for $x = (0, 0)^T$, and why?

3. Prove (7.2.7)–(7.2.9).

4. Suppose that the function $f(x) = \frac{1}{2} \sum_{i=1}^m [r_i(x)]^2$ is twice continuously differentiable, and that the level set

$$\mathcal{L}(x_0) = \{x \mid f(x) \leq f(x_0)\}$$

is bounded. Let the sequence $\{x_k\}$ generated by trust-region type Levenberg-Marquardt Algorithm 7.3.9 converge to x^* with positive definite $\nabla^2 f(x^*)$ and

$$S(x^*) = \sum_{i=1}^m r_i(x^*) \nabla^2 r_i(x^*) = 0.$$

Prove that $\{x_k\}$ converges to x^* with quadratic rate.

5. Let $r \in R^m$, $J \in R^{m \times n}$, $\mu > 0$. Prove that $s = -(J^T J + \mu I)^{-1} J^T r$ is the solution of the least squares problem

$$\min \|Ws + y\|^2,$$

where

$$W = \begin{bmatrix} J \\ \mu^{\frac{1}{2}} I \end{bmatrix}, \quad y = \begin{bmatrix} r \\ 0 \end{bmatrix}.$$

6. Consider nonlinear least-squares problems

$$\min_{x \in R^n} f(x) = \frac{1}{2} r(x)^T r(x) = \frac{1}{2} \sum_{i=1}^m [r_i(x)]^2, \quad m \geq n$$

where $r : R^n \rightarrow R^m$ is a nonlinear function of x and its Jacobian matrix $J(x)$ is full-rank for all $x \in R^n$. Denote the Gauss-Newton direction, the Levenberg-Marquardt direction, and the steepest descent direction respectively by s^{GN} , s^{LM} , and s^C :

$$\begin{aligned} s^{GN} &= -(J^T J)^{-1} J^T r, \\ s^{LM} &= -(J^T J + \mu I)^{-1} J^T r, \\ s^C &= -J^T r. \end{aligned}$$

Prove that

$$\begin{aligned} \lim_{\mu \rightarrow 0} s^{LM}(\mu) &= s^{GN}, \\ \lim_{\mu \rightarrow \infty} \frac{s^{LM}(\mu)}{\|s^{LM}(\mu)\|} &= \frac{s^C}{\|s^C\|}. \end{aligned}$$